

The organization of primary and alternate keys in an Indexed file

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Primary key index

Over the actual data records, the file system builds an index which is used to access the required record

Records in the index are sequenced on ascending primary key.

30 60 99

10 20 30

40 49 60

70 80 99

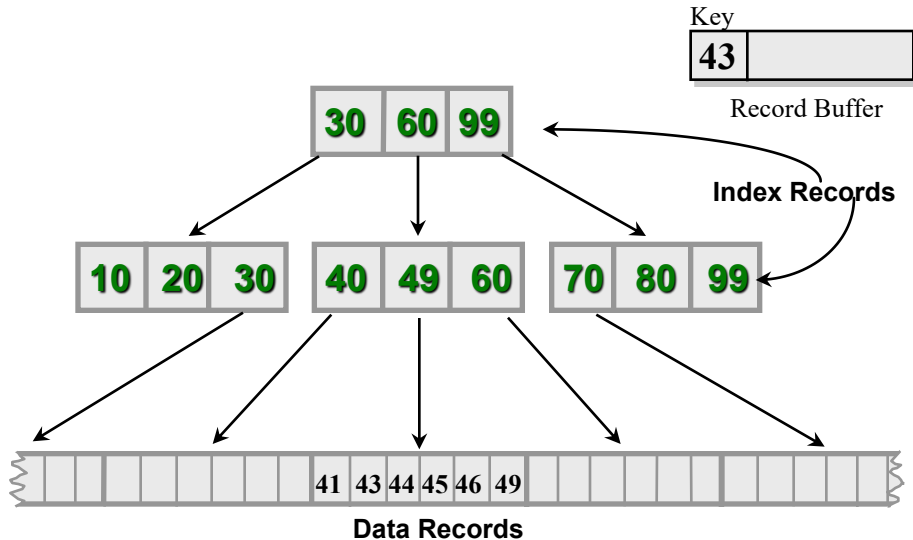
Index Records

41 43 44 45 46 49

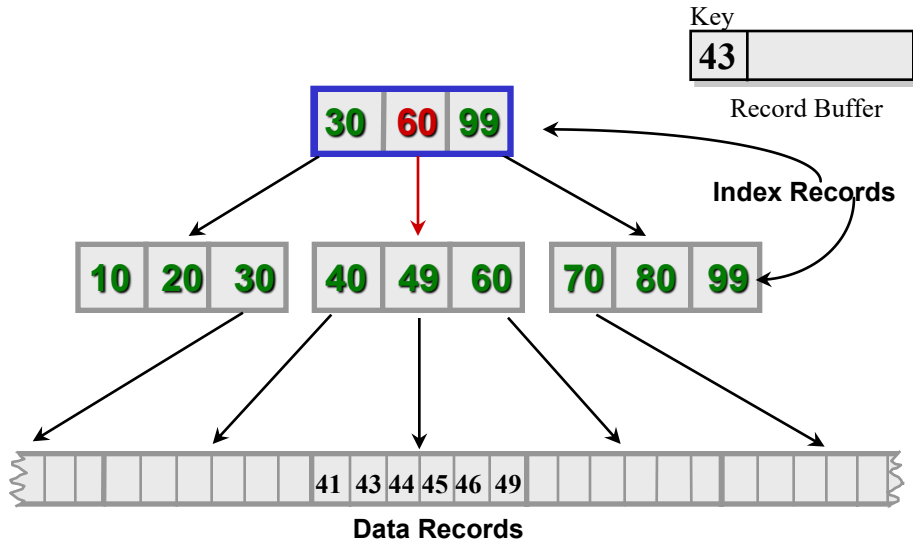
Data Records



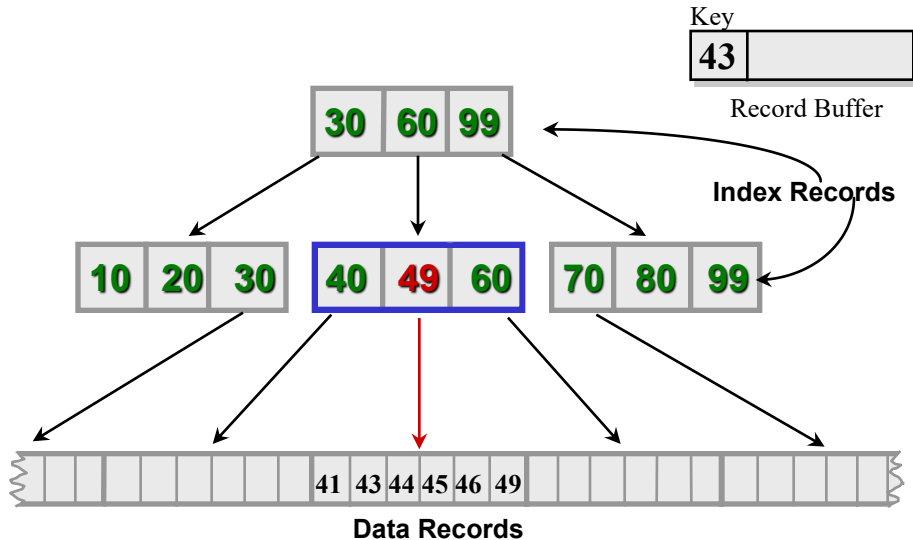
Reading on the primary key



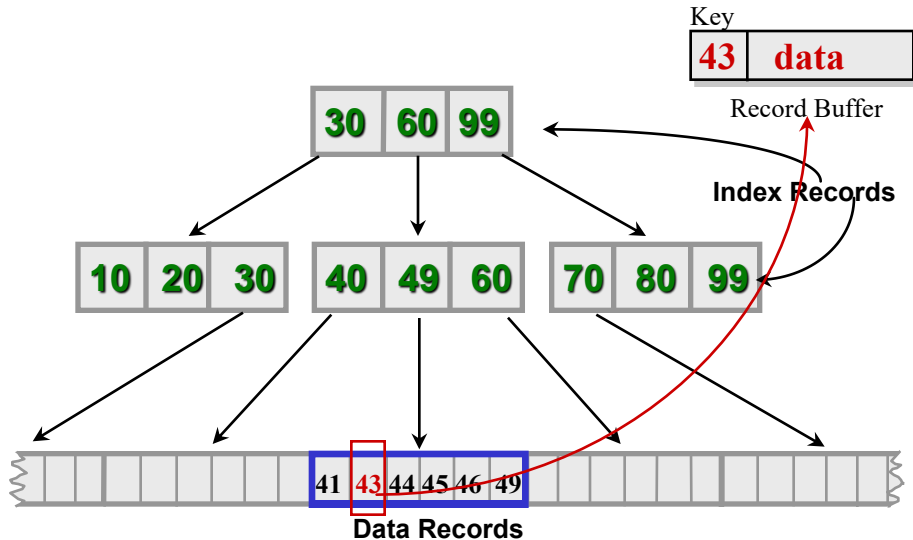
Reading on the primary key



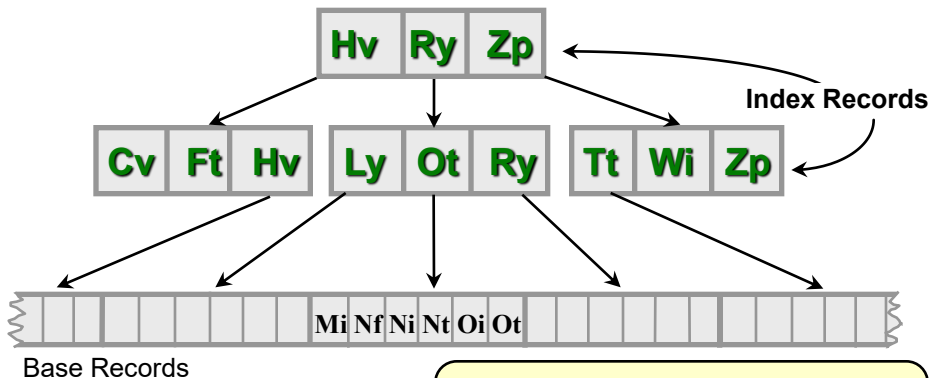
Reading on the primary key



Reading on the primary key

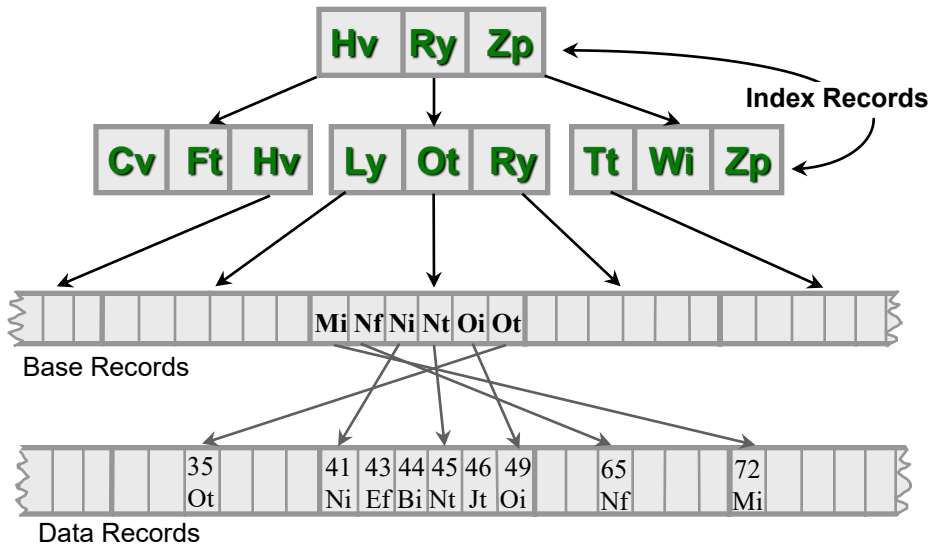


Alternate key index



The base records are organized in ascending alternate key order and this allows us to process the file in ascending alternate key sequence.

Alternate key index



Reading on the alternate key

Pkey Akey

45 **Nt** **data**

Record Buffer

Hv **Ry** **Zp**

Index Records

Cv **Ft** **Hv**

Ly **Ot** **Ry**

Tt **Wi** **Zp**

Mi **Nf** **Ni** **Nt** **Oi** **Ot**

Base Records

35

Ot

41

Ni

43

Ef

44

Bi

45

Nt

46

Jt

49

Oi

65

Nf

72

Mi

Data Records